SEPARATED FINITE TYPE SPACES ARE CLOSED IN SEPARATED FINITE PRESENTATION ONES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3102-3126, following the statement at 3090-3100.

Editorial note. The one-line 'X = lim X_a' follows directly from the ideal colimit and is anchored in the text.

Allowed standard prerequisites (editorial).

- Finite type closed in finite presentation. lemma-finite-type-closed-in-finite-presentation (same chapter, card L26306, cited at ~3103-3105). Closed immersion X -> Z with Z/Y fp.
- Colimit of finite type ideals. lemma-directed-colimit-finite-type (same chapter, cited at ~3108-3111). I = colim I_a over finite type quasi-coherent ideals.
- Closed immersion of finite presentation. Morphisms, Lemma morphisms-lemma-closed-immersion-finite-presentation (cross-chapter, cited at ~3120-3121). X_a -> Z is fp.
- Eventually separated. lemma-eventually-separated (same chapter, cited at ~3123-3125). X_a -> Y separated for some a, since X -> Y is separated.

Source reference. Original lemma 0829: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2519–2519. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 086X: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1541–1541. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0872: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 3036–3036. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01TV: focused-seeds/geometry/S9-spaces-limits-separated-closed-in-finite-presentation/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input;

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the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Proposition 0.1. *Let S be a scheme. $f : X \to Y$ be a morphism of algebraic spaces over S. Assume*

(1) *f is of finite type and separated, and*
(2) *Y is quasi-compact and quasi-separated.*

Then there exists a separated morphism of finite presentation $f' : X' \to Y$ and a closed immersion $X \to X'$ over Y.

Proof. By Lemma 0872 there is a closed immersion $X \to Z$ with Z/Y of finite presentation. Let $\mathcal{I} \subset \mathcal{O}_Z$ be the quasi-coherent sheaf of ideals defining X as a closed subscheme of Y. By Lemma 0829 we can write $\mathcal{I}$ as a directed colimit $\mathcal{I} = \operatorname{colim}_{a \in A} \mathcal{I}_a$ of its quasi-coherent sheaves of ideals of finite type. Let $X_a \subset Z$ be the closed subspace defined by $\mathcal{I}_a$. These form an inverse system indexed by A. The transition morphisms $X_a \to X_{a'}$ are affine because they are closed immersions. Each X_a is quasi-compact and quasi-separated since it is a closed subspace of Z and Z is quasi-compact and quasi-separated by our assumptions. We have $X = \lim_a X_a$ as follows directly from the fact that $\mathcal{I} = \operatorname{colim}_{a \in A} \mathcal{I}_a$. Each of the morphisms $X_a \to Z$ is of finite presentation, see Morphisms, Lemma 01TV. Hence the morphisms $X_a \to Y$ are of finite presentation. Thus it suffices to show that $X_a \to Y$ is separated for some $a \in A$. This follows from Lemma 086X as we have assumed that $X \to Y$ is separated. □